

This is the main coding phase.

## Workout Generation

Function:

`generate_workout_gemini()`

File:

`gemini_generator.py`

It uses user goal and intensity to generate a 7-day plan containing:

- Warm-up
- Main workout
- Exercises
- Sets/repetitions
- Cooldown/recovery tip

## Nutrition Generation

Function:

`generate_nutrition_tip_with_flash()`

File:

`gemini_flash_generator.py`

Gemini Flash generates concise nutrition/recovery advice.

## Feedback-Based Updating

Function:

`update_workout_plan()`

File:

`updated_plan.py`

Example feedback:

Add yoga

Include more cardio

The original plan + feedback are sent to Gemini Pro, which generates the revised plan.

## Database

File:

database.py

Functions include:

save\_user()

save\_plan()

update\_plan()

get\_original\_plan()

get\_user()

User information and generated plans are stored using SQLite + SQLAlchemy.